INCH-POUND
MIL-DTL-16377/49B(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/49A(SHIPS)
18 May 1973

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; EXTENSION, SOLID-STATE, 115 VOLTS, 60 HERTZ, EXPLOSIONPROOF AND WATERTIGHT, DETAIL LIGHTING, SHIPBOARD USE, SYMBOL 306.2SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 4 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.



FIGURE 1. Typical light extension, solid-state, symbol 306.2SSL.

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

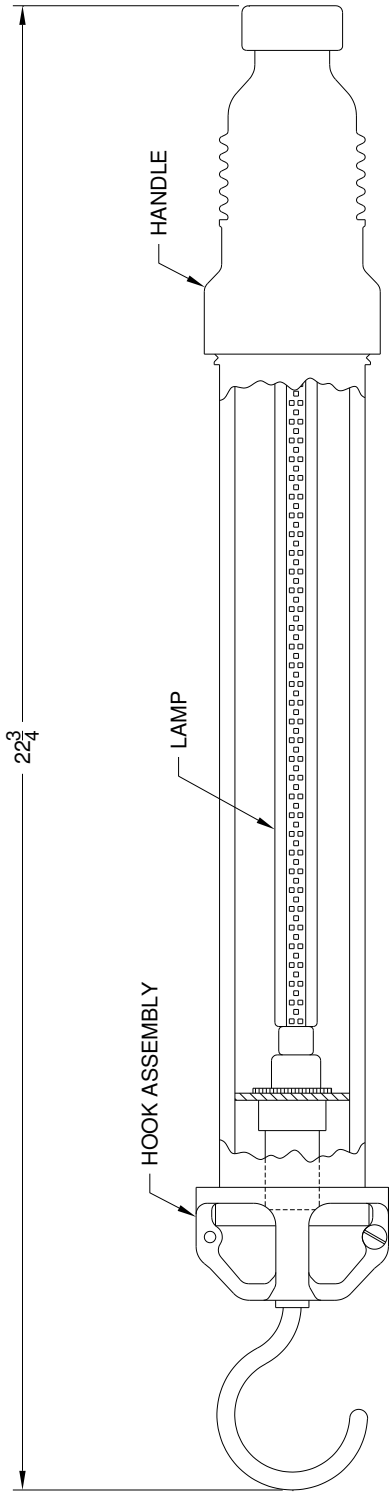
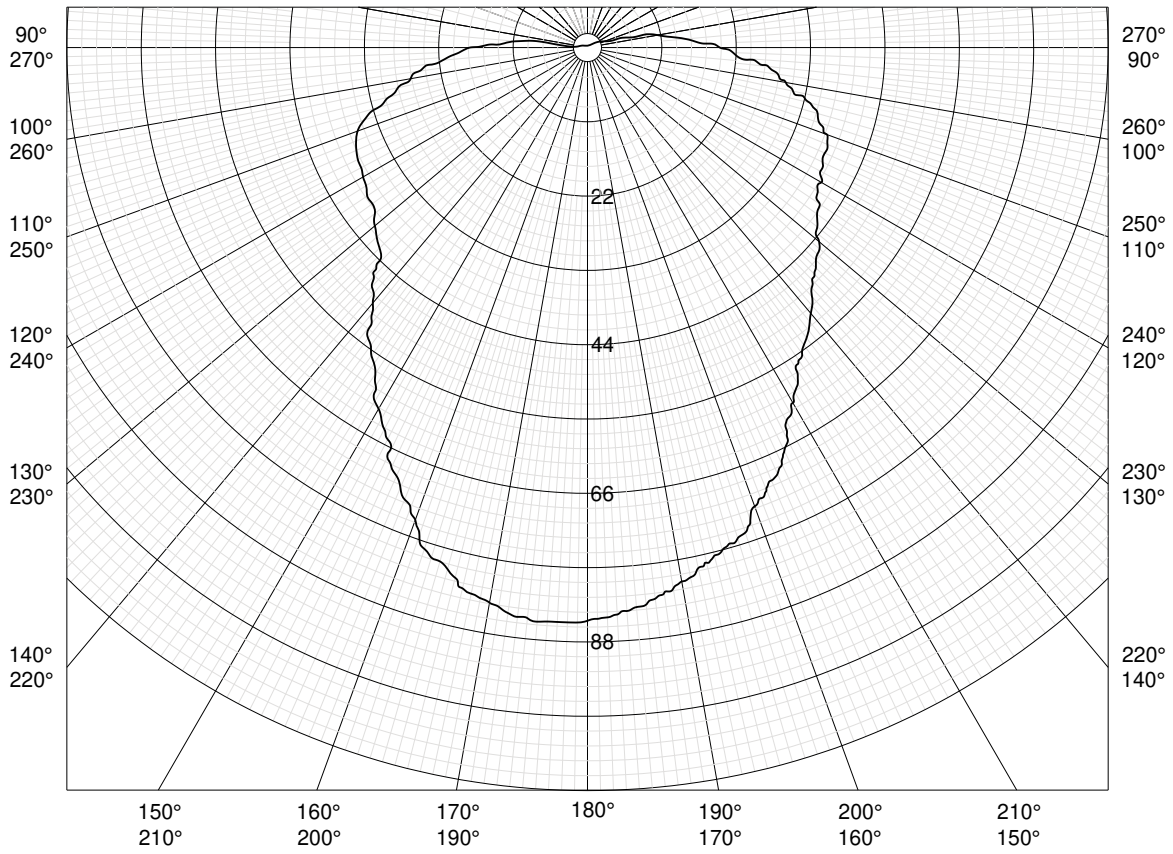


FIGURE 2. Typical section views of light extension, symbol 306.2SSL.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.



NOTES:

1. The plane of this figure shall be perpendicular to the length of the extension light.
2. Distance is 10 feet.

FIGURE 3. Candlepower distribution curve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

CLASSIFICATION

Type: III
Class: 1

SYMBOL NUMBER

306.2SSL: Light, extension, solid-state, 115 volts, 60 hertz, white, explosionproof, and watertight.

DESIGN

Electrical rating:	8 watts maximum, 115 volts, 60 hertz.
Dimensions and configuration:	See figure 2.
Metals:	All exposed metal parts shall be of nonsparking metal or the extension light shall be double insulated in accordance with UL standards.
Enclosure:	The light shall be designed to be explosionproof and watertight and shall be so constructed that no parts will work loose in service. The enclosure shall be capable of withstanding the internal pressures resulting from explosions without bursting or loosening of the joints. The enclosure shall withstand, without rupture or permanent distortion, an internal hydrostatic test. The light shall be designed to be suitable for class I, group D hazardous locations as defined by the National Electrical Code.
Cable:	The light shall be furnished with 50 feet of three-conductor cable, type SO-16/3 or ST-16/3, in accordance with UL 62, if the extension light has exposed metal parts. The third conductor shall be used for grounding. If the extension light is double insulated in accordance with UL standards, the ground conductor is not required, and a two-conductor cable shall be provided.
Ambient temperature:	The light shall be capable of operating in an ambient temperature of 135 °F for 90 hours without damage or loosening of parts.
Weight:	With cable – 10 pounds maximum. Without cable – 3 pounds maximum.
Labels:	The following labels shall be furnished in accordance with MIL-DTL-16377: (a) Type I (one required). (b) Type V, except that the height of the characters shall be not less than 1/16 inch. The information to be shown on this label shall be as follows: "CAUTION Disconnect from the circuit before any maintenance that requires entry into the extension light. Keep all joints tight when in operation. If the connection to the electrical circuit is in an area where explosive gases are present, an explosionproof plug and matching receptacle shall be used. If the electrical connection is made outside of the hazardous area, a three-conductor grounded plug may be used. The light shall not be disassembled in a hazardous location." (c) A warning label with red letters on white background with the following information: "EXPLOSIONPROOF AND WATERTIGHT EXTENSION LIGHT FOR HAZARDOUS LOCATIONS CLASS 1, GROUP D. MAXIMUM OPERATING SURFACE TEMPERATURE: 50 °C

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

WARNING
FOR USE IN CLASS 1, GROUP D ATMOSPHERES ONLY, IN WHICH, AMONG OTHERS, THE FOLLOWING ARE INCLUDED: GASOLINE, PETROLEUM NAPHTHA, ALCOHOL, ACETONE, LACQUER SOLVENT VAPORS, AND NATURAL GAS."

Photometrics:	See figure 3.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.6.17 of MIL-DTL-16377.

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table I.
Conformance inspection:	
Comparison inspection:	Same as the first article inspection.
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION

This light is intended to be used as a general-purpose extension light for detail illumination onboard ship and as a work light in areas where explosive gases may be present.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

TABLE I. First article and comparison inspection for complete fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377 or Reference to UL 844	Test Paragraph of MIL-DTL-16377 or Reference to UL 844	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Rough usage	UL 844	UL 844	Rough-usage test
Drop	UL 844	UL 844	Drop test
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Ambient temperature	1/	1/	---
Grounding circuit	3.6.13.3	4.8.11	2/
Continuity of grounding	3.6.13.4	4.8.12	2/
Leakage current	3.6.13.5	4.8.13	---
Optical uniformity	3.7.4.1.1	4.8.4	---
Stress relief	3.8.5.3	4.8.5	---
Cable pull	UL 844	UL 844	Strain-relief test
Magnetic permeability	3.4.8	4.8.18	---
Salt fog	3.3.15	4.8.10	---
Enclosure effectiveness	3.5.11	4.8.15	Explosionproof
Hydrostatic pressure	3.5.11	4.8.15.2 1/	---
Photometrics	3.7.1	4.8.6	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature, white light only	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Optical uniformity	3.7.6.1.1.1	4.8.4	---
Stress relief	3.8.5	4.8.5	---

NOTES:

1/ As specified herein.

2/ Only required for extension lights with exposed metal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.

TABLE II. Part number and general information.

Military Part Number M16377/49-	Item Name	National Stock Number (NSN)
306.2SSL	Complete light, symbol 306.2SSL	6230-00-244-3996

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6230-2023-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:49Z
Check the source to verify that this is the current version before use.